MATH 152 **EXAM III** NAME _____

Mrs. Bonny Tighe PART ONE

11.8-10.1 TA_____ Wed. 11/28/2018

50 points

1. (4 points each) Circle the letter for the correct answer

a) The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n (\pi)^{2n+1}}{9^n (2n+1)!}$

A) $3/2$

B) $3\sqrt{3}/2$

C) $\sqrt{3}/6$

D) $\sin(1/2)$

E) Some other answer

b) The interval of convergence for the series $\sum_{n=0}^{\infty} \frac{(x-1)^n}{n^2+2}$

A) $(0,2)$

B) $(0,2]$

C) $[0,2]$

D) $[0,2)$

E) Some other answer

c) The Maclaurin series representation for $f(x) = \tan^{-1}(3x^2)$ is:

A) $\sum_{n=0}^{\infty} \frac{(-1)^n 3x^{4n+2}}{2n}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n} x^{2n+1}}{2n++1}$

C) $\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1} x^{4n+2}}{(2n+1)!}$

D) $\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1} x^{4n+2}}{2n+1}$

E) Some other answer

d) Which of the following parabola has its vertex at (2, 0) and the directrix of $x = 1$

A) $(x-2)^2 = 4y$

B) $(x-2)^2 = -4y$

C) $y^2 = 4(x-2)$

D) $y^2 = -4(x-2)$
E) Some other answer

e) The series which is equivalent to $\sum_{n=1}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ is:

A) $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n)!}$
B) $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n+2}}{(2n+1)!}$
C) $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n+2}}{(2n+2)!}$
D) $\sum_{n=0}^{\infty} \frac{(-1)^{n-1} x^{2n}}{(2n-1)!}$
E) Some other answer

f) The derivative of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ is

A) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$

B) $\sum_{n=0}^{\infty} \frac{(-1)^n (2n) x^{2n}}{(2n)!}$
C) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n)!}$
D) $\sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n}}{(2n)!}$
E) Some other answer

g) $T_6(x)$ for $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{4^{n+1}}$ is

A) $\frac{1}{4} - \frac{1}{4^2}x^2 + \frac{1}{4^3}x^4 - \frac{1}{4^4}x^6 + \frac{1}{4^5}x^8 - \frac{1}{4^6}x^{10}$
B) $\frac{1}{4} - \frac{1}{4^2} + \frac{1}{4^3} - \frac{1}{4^4} + \frac{1}{4^5} - \frac{1}{4^6}$
C) $\frac{1}{4} + \frac{1}{4^2}x^2 + \frac{1}{4^3}x^4 + \frac{1}{4^4}x^6$

D) $\frac{1}{4} - \frac{1}{4^2}x^2 + \frac{1}{4^3}x^4 - \frac{1}{4^4}x^6$

E) Some other answer

2. (2 points each) True or False

a) TRUE FALSE $4 \cdot 5 \cdot 6 \cdot 7 \cdot 8 = \frac{8!}{3!}$

b) TRUE FALSE The Maclaurin series for $\sin x = \sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^{2n}}{(2n)!}$

c) TRUE FALSE The asymptotes for the hyperbola $\frac{(y-1)^2}{4} - \frac{(x+2)^2}{9} = 1$ are $y = \pm\frac{2}{3}x$

d) TRUE FALSE Given $f(2) = 1$, $f'(2) = -2/3$, $f''(2) = 25/3^2$, $f'''(2) = -258/3^3$, etc. then $f^{(n)}(2) = \frac{(-1)^n (2 \cdot 5 \cdot 8 \cdots (3n-1)}{3^n}$

e) TRUE FALSE If $-2 < x < 4$ then $|x-2| < 2$

f) TRUE FALSE $\frac{1}{x-1} = \sum_{n=1}^{\infty} \frac{1}{x}\left(\frac{1}{x}\right)^n$

g) TRUE FALSE The radius of convergence for the series $\sum_{n=0}^{\infty} \frac{(-1)^{n+1} x^n}{3^{2n}}$ is 3.

h) TRUE FALSE $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!} = \frac{1}{e}$

i) TRUE FALSE The sum of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$ is 1.

3. (9 points) Use the Binomial Theorem to find the function as a power series.

$f(x) = \frac{1}{\sqrt{1-x^3}}$

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1. (4 points) Find the vertex, focus, and directrix. Sketch the graph

$y^2 + 4y + 2x + 2 = 0$

2. (4 points) Find the vertices, foci, asymptotes, and center. Sketch the graph.
$9(x+2)^2 - 4(y-1)^2 = 36$

3. (4 points) Find the equation of the ellipse with vertex at (-2, 3) and foci at (-2, 5) and (-2, 7)

4. (5 points) Sketch the plane curve defined by the given parametric equation. Eliminate the parameter to find a Cartesian equation of the curve. Indicate with an arrow the direction in which the curve is traced as t increases. $x = 2\sin\theta, \quad y = 3\cos\theta$

5. (5 points) Use series to evaluate the limit $\lim_{x \to 0} \frac{1 - \cos(x^2)}{e^{x^2} - 1 - x^2}$

86 (10 points) Find the Taylor series for $f(x) = \frac{1}{(3+x)^3}$ centered at a = 2

7. (5 points) Find the Interval and Radius of convergence for the power series

$$\sum_{n=0}^{\infty} \frac{(-1)^n (x-2)^n}{3^n (n+2)}$$

8. (5 points) a) Find a Taylor polynomial of degree 3 for $f(x) = \sqrt{x}$ centered at $a = 9$.

b) (4 points) Use it to approximate $\sqrt{9.1}$.

c) (4 points) How accurate is this approximation when $|x - 9| < 0.5$

If $\left|f^{(n+1)}(x)\right| \le M$ for $|x - a| \le d$, then the remainder $R_n(x) \le \frac{M}{(n+1)!}|x - a|^{n+1}$ for $|x - a| \le d$

9. (5 points) Evaluate the integral as a power series and approximate to T_6.

$$\int_0^{0.5} \frac{x}{4 - x^2}\, dx$$